BMAS XXX: NETWORK SECURITY

Course Objectives: The aim of the course is to build knowledge and understanding of Cryptography, mutual trust, Transport level security; IP Security; Wireless Network security; network endpoint security; IoT protocol security

Credits: 04 **L–T–P: 3–1–0**

Module No.	Contents	Hours (Approx.)
I	**Cryptography & Trust Management:** Symmetric vs. Asymmetric Cryptography, Hash Functions (MD5, SHA), AES, RSA, SSL/TLS Overview, Public Key Infrastructure (PKI), Digital Signatures, Certificate Authorities (CAs), Cryptographic Key Management & Distribution, Kerberos, X.509 Certificates. **Network Security & Protocols:** CIA Triad, Services & Ports, TCP/IP Communication Flags, Port Scanning with Nmap, Attacks Across Network Layers, Link Layer, Network Layer, Transport Layer & Application Layer Security Countermeasures, Network Access Control. **IP & Transport Security:** IP Security Overview, IPsec Components (AH, ESP, IKE), Transport & Tunnel Modes, IP Security Policies, TLS Architecture, TLS Record Protocol, TLS Handshake, TLS Certificates, TLS Vulnerabilities, HTTPS, SSH. **Security Policies & Countermeasures:** Mutual Trust Mechanisms, Network & Transport-Level Security Measures, IP Security & TLS Countermeasures. **Email Security:** Phishing & BEC, Email Authentication (SPF, DKIM, DMARC, PGP), Secure Email Protocols (SMTP, IMAP, POP3, STARTTLS), Malware & Ransomware, MFA & Best Practices.	20
II	**Network Vulnerabilities:** Physical & Data Link Attacks (Cable Tapping, MAC Flooding, ARP Spoofing), Network & Transport Threats (IP Spoofing, ICMP Attacks, SYN/UDP Flooding, Port Scanning), Session & Application Exploits (Session Hijacking, MITM, SQL Injection, DoS/DDoS, Botnets), Protocol & DNS Attacks (DNS Spoofing), Network Access Control (NAC), Proxy Server, and Countermeasures. **End-Point Security:** Firewall Architecture & Types, Firewall Rules & Policies, DMZ, (iptables, UFW, pfSense), Bypassing, Intrusion Detection & Prevention Systems (IDS/IPS) with Detection Techniques (Signature, Anomaly, Behavioral), and Countermeasures. **VPN:** Functions of a VPN, types of VPN, VPN tunneling protocol (PPTP, L2TP, SSTP, OpenVPN, WireGuard), SSH Tunnelling, VPN Topologies, VPN concerns, SOCKS Proxy, countermeasures **Wireless Security**: Wireless terminology, Wi-Fi standards (IEEE 802.11, 802.11a, 802.11b, etc), Wi-Fi authentication process (PSK mode, centralized mode), Wireless (EAP, TKIP, AES, CCMP, WEP, WPA, WPA2, WPA3, etc), countermeasures **IoT Protocol Security**: IoT Architecture and Taxonomy, IoT Network and Communication Authentication, Attacks on IoT infrastructure, Protection Mechanisms for IoT	20

Text Books:

Attacking Network Protocols (A hacker's guide to capture, analysis and exploitation) James Forshaw, No Starch Press

Reference Books:

- The Network Security Test Lab - A Step-by-Step Guide Michael Gregg, Wiley
- Network Security Essentials: Applications and Standards" by William Stallings
- Understanding Network Hacks (Attack and Defense with Python 3) Bastian Ballmann, Springer
- Linux Firewalls & Iptables – Michael Rash (or) Mastering pfSense – David Zientara
- Network Security with OpenSSL" by John Viega, Matt Messier, and Pravir Chandra
- Practical Network Security: Tools and Techniques for Securing Your Network" by Jeremy Faircloth

Focus: This course focuses on employability and skill development aligned with all CO's.

Learning Outcomes: After completion of the course, student will be able to:

CO1: Understand and Differentiate Cryptographic Techniques
CO2: Implement and Evaluate Secure Communication Mechanisms
CO3: Analyze Network Protocols and Security Mechanisms
CO4: Apply IP and Transport-Level Security
CO5: Evaluate and Mitigate Email and Network Vulnerabilities
CO6: Implement End-Point Security Solutions
CO7: Design Secure VPN and Wireless Architectures

COs	POs/PSOs
CO1	PO1, PO2 / PSO1
CO2	PO2, PO3, PO4 / PSO2
CO3	PO1, PO3, PO5 / PSO3
CO4	PO2, PO4, PO6 / PSO4
CO5	PO3, PO5, PO7 / PSO2, PSO4
CO6	PO4, PO5, PO6 / PSO3
CO7	PO3, PO6, PO8 / PSO4